

Gabriel Cruz

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EDUCATION

Georgia Institute of Technology

B.S. in Computer Science and Mathematics, GPA: 3.2

Atlanta, GA

Aug 2024 - May 2028

- Threads: Systems & Architecture, Intelligence
- Coursework: Computer Architecture, Digital Design Lab, Systems & Networks, Data Structures & Algorithms, Design & Analysis of Algorithms

EXPERIENCE

Electrical Engineering Intern

June 2023 - May 2024

Johns Hopkins Applied Physics Laboratory

Laurel, MD

- Led small teams building prototypes for wireless power transfer (WPT) and electrical impedance spectroscopy (EIS), including a low-cost 2.45 GHz closed-loop WPT system with adaptive phase and impedance tuning
- Researched microwave signals for defense and utility applications; simulated 2.45 GHz and 5.8 GHz ISM signals in GNU Radio and MATLAB/Simulink, reducing energy rebound by 30%
- Developed a predictive Free Space Path Loss model and tested existing systems to improve wireless system design accuracy
- Built an automated leaf health monitoring pipeline using an Arduino and Python (Plotly) that counted xylem in leaves and flagged whether they fell in a healthy range, achieving 99.1% detection accuracy

PROJECTS

WayFinder Smart Cane | C++, ESP32, FreeRTOS, Arduino

- Developed firmware for an assistive cane prototype integrating two ultrasonic sensors, proximity-based audio alerts, and a five-second emergency button
- Alternated sensor measurements at 80 ms intervals to reduce interference; prioritized the closest qualifying obstacle using independent distance thresholds and stale-reading rejection
- Implemented Telegram SOS messaging over TLS in a separate FreeRTOS task so sensing could continue during network requests; added automatic Wi-Fi reconnection and button debouncing

LC-5200 Microcoded Processor | CircuitSim, Microcode, Assembly

- Built a 32-bit microcoded datapath (ALU, register file, control unit) and added interrupt support, including EI, DI, IN, and RETI instructions and an assembly interrupt handler
- Achieved a 1.64× average speedup across three benchmarks using a dual-ported register file, an immediate-to-ALU bypass, and a dedicated PC incrementer that cut fetch from 3 cycles to 2
- Diagnosed and fixed an incorrect ALU function in the BEQ microcode by tracing ALU inputs and outputs

Multithreaded CPU Scheduler | C, pthreads, Linux

- Implemented FCFS, Round Robin, SRTF, and preemptive priority schedulers for a multiprocessor OS simulator with one thread per simulated CPU
- Protected shared ready queues and per-CPU state with mutexes and condition variables, synchronizing access and blocking idle CPUs

Cache-Blocked Matrix Multiplication | C++

- Built a templated Matrix<T> class with row-major layout and naive and cache-blocked multiplication
- Benchmarked with a custom GFLOPs/sec harness, achieving a 6.3× speedup (3.0% to 19.1% of theoretical peak) and diagnosing poor cache reuse as the bottleneck

Signal Logic Synthesizer | VHDL, Quartus, DE10 FPGA

- Designed and implemented a signal logic synthesizer in VHDL, targeting the DE10 FPGA board
- Verified the design in simulation and on hardware, using an oscilloscope to check output signals

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Assembly, Java, MATLAB

Hardware Description Language: VHDL

Embedded: ESP32, FreeRTOS, Arduino framework, GPIO, interrupts, state machines

Tools: Linux, Git, Make, GDB, pthreads, MATLAB/Simulink, Quartus, GNU Radio

Hardware: Arduino, DE10 FPGA board, oscilloscope, RF/microwave